Design of a Reproducible Healthy Aging Policy Observatory: Infrastructure for Trustworthy Aging-Policy Evidence in Canada

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Abstract

Healthy aging is among the defining public-policy challenges of aging societies, and governments act on it continually. Yet the evidence base for *whether those actions work* suffers less from a shortage of analysis than from a shortage of *trustworthy infrastructure*: policy text and outcome data live apart, published numbers rarely trace to their sources, and descriptive co-movement is routinely dressed as causation. **We argue these are infrastructure problems, not analysis problems** — and that the response is a reproducible research instrument, an *observatory*, rather than another dashboard. We frame the design around a conceptual model, the **Healthy Aging Policy Intelligence Cycle** (Observation → Evidence → Indicator → Policy → Outcome → Feedback) — a policy-level instance of a Learning Health System — and ask four design-evaluation questions: can every published number be traced to its exact source (provenance); does the pipeline reproduce identical results (reproducibility); can the AI layer emit an ungrounded citation (grounding); and are composite comparisons robust to method choices (robustness). We present the Canadian Healthy Aging Policy Observatory, whose data model makes provenance structural (immutable observations bound to versioned, content-addressed dataset snapshots) and whose analytics enforce an association-versus-causation distinction in software. Instantiated on Nova Scotia and the federal level with a Care Access focus — an initial implementation, with scaling across provinces as future work the architecture enables — the instrument answers the first three design questions affirmatively: 100% of published values resolve to an upstream source, the pipeline reproduces deterministically from versioned inputs, and the assistant cannot cite outside a fixed evidence pack. The fourth it answers with the caution it is built to give: composite levels move by at most 2.3 points across three weighting schemes, but the Nova Scotia–federal ordering does not hold under all three, so the instrument reports the two as indistinguishable rather than ranking them. **The design itself is the contribution**: a reproducible, provenance-first observatory is a durable paradigm for aging-policy evidence that compounds across a multi-paper program, in which the index methodology, the grounded-AI framework, and the causal-evaluation studies are each deepened in their own right.

Keywords: healthy aging, health informatics, learning health system, policy evaluation, reproducible research infrastructure, data provenance, composite indicators, interrupted time series, retrieval-augmented generation, Nova Scotia.

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1. Introduction

1.1 The problem is infrastructure, not analysis

Canada, like most high-income countries, is aging quickly, and governments at every level respond with policy — home-care investment, long-term-care (LTC) capacity, dementia strategies, seniors' income supports, digital-inclusion programs. Yet the machinery for *evaluating* those policies lags the machinery for *announcing* them. Three gaps recur:

1. **Fragmentation.** Policy text, budgets, and target populations live in press releases and PDFs; outcome data live in separate agencies (Statistics Canada, the Canadian Institute for Health Information, provincial open-data portals). Nothing links a policy to the indicators it was meant to move.
2. **Irreproducibility.** Published dashboards and reports rarely let a reader trace a headline number to the specific upstream table, vintage, and transformation that produced it; re-running an analysis a year later often yields a different number for unstated reasons.
3. **Causal overreach.** Descriptive co-movement ("spending rose and ED visits fell") is routinely presented in language implying causation, without the design or assumptions that would justify it.

Our central claim is that **these are infrastructure problems, not analysis problems.** No new estimator fixes a number that cannot be traced to its source, an analysis that cannot be reproduced, or a claim whose causal status is ambiguous by default. What is missing is a standing, reproducible *instrument* — an **observatory** — that continuously ingests policies and outcomes, quantifies them with a documented and versioned method, and makes honest inference the path of least resistance. The distinction is the one astronomy draws between a telescope and a press photo: the value is in the instrument's reliability and in the fact that others can point it themselves.

1.2 Why an aging-policy observatory is health informatics

Policy is an upstream determinant of health: home-care funding, LTC staffing, and income supports shape the functional ability, independence, and outcomes of older adults as surely as any bedside intervention. Health informatics has correspondingly broadened from clinical decision support toward **population- and policy-level** decision support and **knowledge translation** — turning evidence into actionable, traceable syntheses for decision-makers. The observatory is precisely such an instrument, and its design mirrors a core health-informatics idea: the **Learning Health System** (LHS) [11], in which data generate knowledge that improves practice and thereby generates new data, in a continuous loop. Where a clinical LHS closes that loop around care delivery, the observatory closes it around *policy* — an LHS for healthy-aging governance. An observatory is, moreover, a **socio-technical system**: it couples data and AI infrastructure with the researchers, agencies, and governments whose practices it must fit, so its hardest design challenges are as much organizational — trust, provenance, honest communication — as computational. Reading the design through evidence, knowledge translation, and decision support is what makes it a health-informatics contribution rather than a software-engineering one.

1.3 Research questions

Because the contribution is a design, we evaluate it as a design. We ask four **design-evaluation questions** whose answers are properties of the artifact, not outcomes of a clinical trial:

- **RQ1 (Provenance).** Can every published quantitative value be traced to the exact upstream source it came from?
- **RQ2 (Reproducibility).** Does the pipeline reproduce identical results from the same inputs, and is ingestion idempotent under unchanged upstream data?
- **RQ3 (Grounding).** Can the AI layer emit a factual claim or citation that is not grounded in a fixed, visible evidence set?
- **RQ4 (Robustness).** Are the composite index's headline comparisons robust to its most contestable method choice, the domain weighting?

§7 answers each with evidence from the instantiated instrument.

1.4 Contribution

The primary contribution of this paper is **the design of a reproducible healthy-aging policy observatory, together with the conceptual framework it realizes** — the Healthy Aging Policy Intelligence Cycle (§3). The design's defining commitments are provenance as a structural property, reproducibility by construction, and an association/causation discipline enforced in software. The observatory *instantiates* three modules — a documented composite index (HAPI), a quasi-experimental analytics layer, and a grounded AI research assistant — but these are presented here as evidence that the framework runs end-to-end, not as the paper's contribution; each is the subject of its own paper in the program (§8). This scoping is deliberate: a single, well-evaluated design claim is worth more than an inventory of subsystems.

2. Related work

Learning health systems and health-informatics infrastructure. The LHS concept [11] frames continuous improvement as a data→knowledge→practice→data cycle; most LHS work targets clinical care. We extend the idea *upstream* to policy, and we take seriously the informatics infrastructure — provenance, versioning, reproducibility — that an LHS presupposes but often under-specifies. Our data-stewardship commitments follow the FAIR principles [9].

Composite indices of aging. The WHO *World Report on Ageing and Health* [1] and the *Decade of Healthy Ageing* [2] articulate intrinsic capacity and functional ability; the European Active Ageing Index [5] and HelpAge's Global AgeWatch Index [4] operationalize domains into composite scores. HAPI is in this tradition but is policy-facing, reproducible by construction, and transparent about its own weighting; its methodology is developed in full in Paper 2 (§8) and appears here only as an instantiated module. We follow OECD/JRC guidance [3] on transparency and sensitivity analysis.

Policy monitors and quasi-experimental evaluation. Established health-policy observatories — the European Observatory on Health Systems and Policies, and the monitoring functions of the OECD and WHO — synthesize cross-country policy and produce authoritative reports, but they are primarily *reporting* instruments: they seldom expose per-value provenance, reproduce end-to-end from source, or attach a formal causal discipline to their comparisons. Government/NGO trackers likewise catalog policies without integrating outcomes or supporting inference; academic evaluations produce rigorous single-policy studies but rarely as reusable infrastructure. The segmented-regression interrupted-time-series literature [6, 7, 8] provides the reference designs; the observatory operationalizes them under an explicit tagging discipline and defers their empirical application to Paper 4.

LLMs for policy analysis. Retrieval-augmented generation (RAG) [10] makes it feasible to draft

grounded syntheses over a fixed corpus; the risk is fabricated citations and unsupported claims. Our assistant constrains the model to cite *only* items in a pre-assembled evidence pack, making fabrication structurally detectable; the framework is developed in Paper 3.

The gap. These bodies of work each cover one facet and stop at its edge: existing aging observatories and indicator dashboards focus on *reporting* measures; composite-index research focuses on *measurement*; policy-evaluation studies focus on *single interventions*; and Learning Health System work targets primarily *clinical care*. To our knowledge, no existing system integrates provenance-first infrastructure, reproducibility by construction, grounded AI, and an explicit causal discipline into a *unified* observatory for healthy-aging policy. That integration — not any single component — is the gap this design fills.

3. Conceptual framework: the Healthy Aging Policy Intelligence Cycle

The design is organized by a conceptual model that doubles as its theory of value. Healthy-aging policy intelligence is a cycle of six stages (Figure 1):

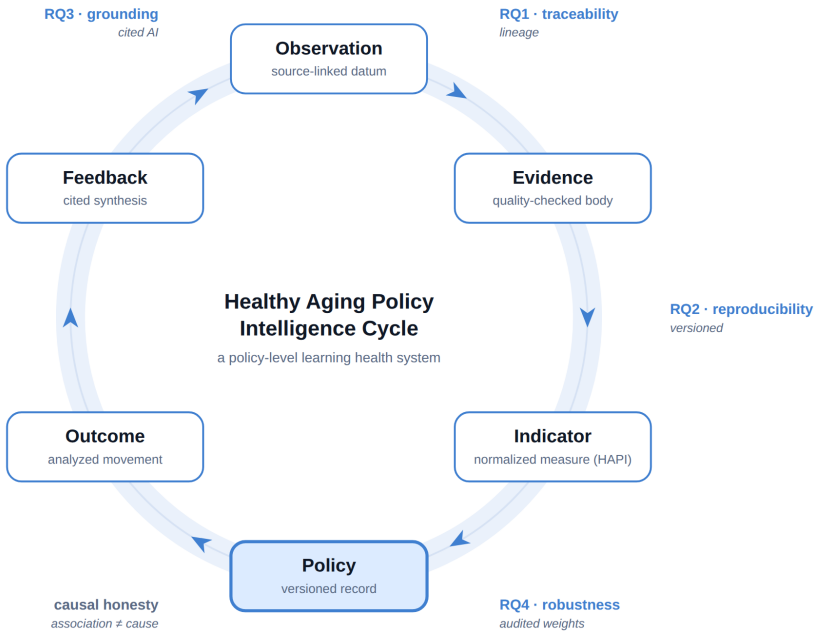


Figure 1. The Healthy Aging Policy Intelligence Cycle. Six stages form a closed loop; the outer labels mark where the design hardens trust and which design-evaluation question (§1.3, §7) each answers — RQ1 traceability, RQ2 reproducibility, RQ4 robustness, RQ3 grounding — with the Outcome stage's causal honesty as an additional discipline. Theory (the cycle), design (the modules), and evaluation (the RQs) thus close on one figure.

- **Observation** — an immutable, source-linked datum ingested from an upstream table.
- **Evidence** — observations assembled and quality-checked into an interpretable body, with literature and prior findings.
- **Indicator** — evidence normalized into a documented, comparable measure (HAPI).

- **Policy** — a versioned policy record linked to the indicators it intends to move.
- **Outcome** — the movement of indicators over time, analyzed under an association/causation discipline.
- **Feedback** — findings and syntheses returned to researchers and policymakers, which motivate new observations, closing the loop.

Two properties make this more than a diagram. First, it is a **policy-level Learning Health System** [11]: each turn of the cycle converts data into knowledge that informs policy and thereby generates new data. Second, it names exactly the points where trust is usually lost — the arrows — and the design hardens each: Observation→Evidence is made *traceable* (lineage), Indicator is made *reproducible and version-stamped*, Outcome is made *honest* (causal tagging), and Feedback is made *grounded* (cited AI). The five platform modules (§4) are the machinery that turns this cycle; the rest of the paper shows the machinery exists, runs, and can be trusted.

4. System design

4.1 Modules as cycle stages

The Observatory is five modules over one database and one provenance discipline (Figure 2), each realizing part of the cycle:

1. **Policy Library** (Policy) — a jurisdiction-aware, time-stamped, versioned record of aging policies: department, budget, target population, lifecycle, themes, and AI-assisted summaries.
2. **Data Hub** (Observation → Evidence) — connectors that ingest open data into immutable observations with full lineage and automated quality checks.
3. **Indicators / HAPI** (Indicator) — a documented composite over six domains, computed from observations and stored with a method version.
4. **Policy Analytics** (Outcome) — descriptive trends and quasi-experimental designs under an explicit association/causation tag.
5. **AI Research Assistant** (Feedback) — retrieval over the platform's own stores that produces cited evidence packs and draft reviews.

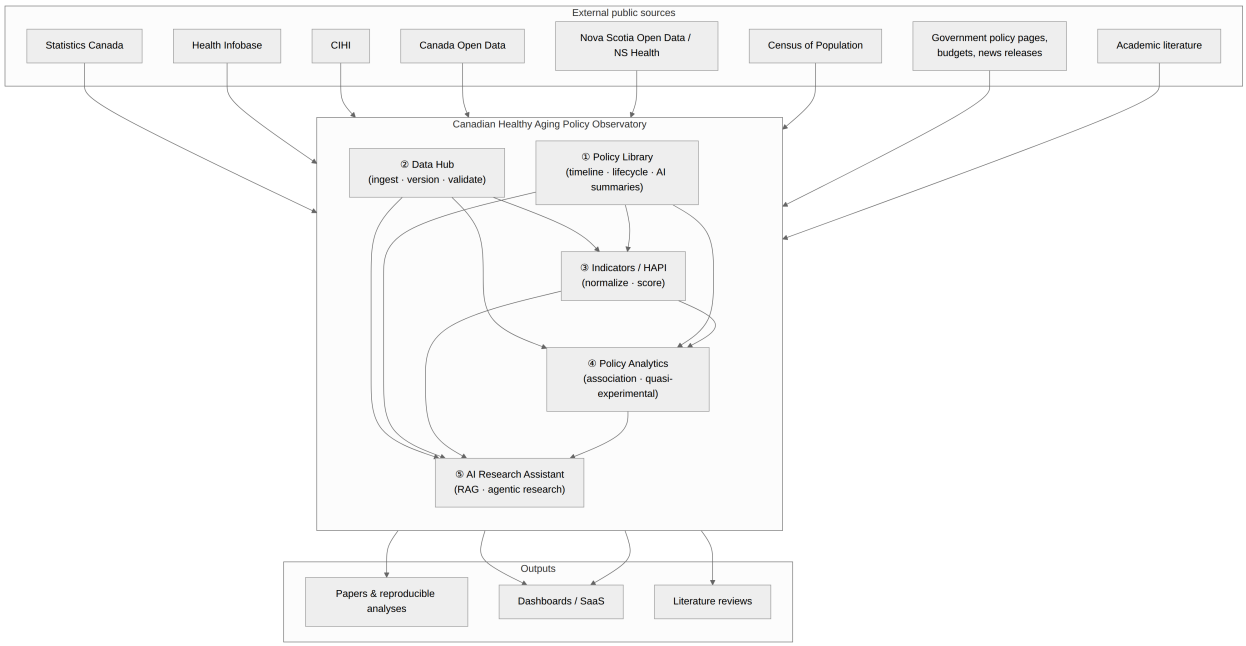


Figure 2. System context: external public sources flow into the five Observatory modules, which produce papers, dashboards, and literature reviews. (Source: `docs/01-platform-overview.md`.)

4.2 Architecture

The platform is a monorepo (Figure 3): a **Next.js/TypeScript** web app (`apps/web`) for the researcher-facing surfaces; a **Python** pipeline (`pipeline/hapi_pipeline`) for ingestion, scoring, and analysis; a **PostgreSQL** schema with forward-only migrations (`db/`); and a shared contracts package. The pipeline is the reproducible core (deterministic transforms, versioned method); the web app is a read-mostly view over what the pipeline produced. A number in the UI is never computed in the browser — it renders a stored, lineage-bearing record. The AI layer wraps a hosted LLM behind narrow, audited entry points, and every generation is logged.

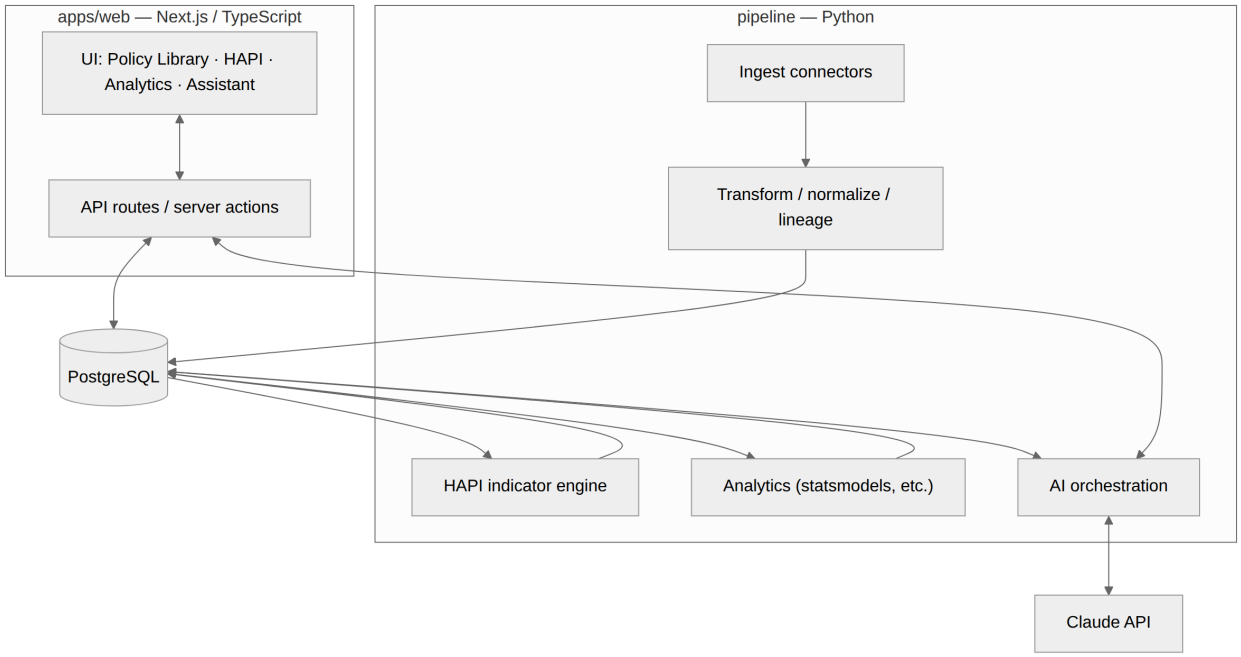


Figure 3. Architecture: the Python pipeline writes to PostgreSQL (raw → transformed → indicators → analytics); the Next.js app reads those tables and invokes the AI layer. (Source: `docs/02-architecture.md`.)

4.3 Design principles

Foundation before breadth (get the model, lineage, and method right on a narrow slice, then widen additively); *every artifact is demonstrable and reproducible*; and *researcher-first* (public/decision-support surfaces follow a credible research core).

5. Data model: making provenance structural

The schema (Figure 4) centers on entities that make provenance and reproducibility *structural* rather than conventional:

- **Jurisdiction** — a hierarchical tree (Canada → Federal / Nova Scotia → …); adding a province is a row, not a schema change.

- **Policy / policy_version** — versioned policy records linked to a jurisdiction.
- **Indicator** — the definition of a measure: meaning, unit, normalization, per-capita rule.
- **DataSource** — upstream provider metadata (agency, licence, cadence, access method).
- **DatasetVersion** — a specific *retrieval*: timestamp + content **checksum**, the unit of idempotency.
- **Observation** — an immutable (jurisdiction × indicator × time) value bound to its DatasetVersion; never mutated — a new vintage yields a new version and new observations.
- **hapi_score** — a computed domain/composite score at a given `method_version`.
- **policy_indicator, indicator_source** — the many-to-many links that make the platform a small knowledge graph.
- **analysis_finding, literature** — analytic results (tier-tagged) and the literature knowledge base.

The key property is the **lineage chain**: every value resolves as `Observation → DatasetVersion → DataSource`. Because observations are immutable and dataset versions are content-addressed, analyses are reproducible and ingestion is idempotent (re-ingesting unchanged upstream data is a no-op). This is what lets the platform claim any published number is auditable to the exact upstream table it came from — and it is what §7 evaluates directly (RQ1, RQ2).

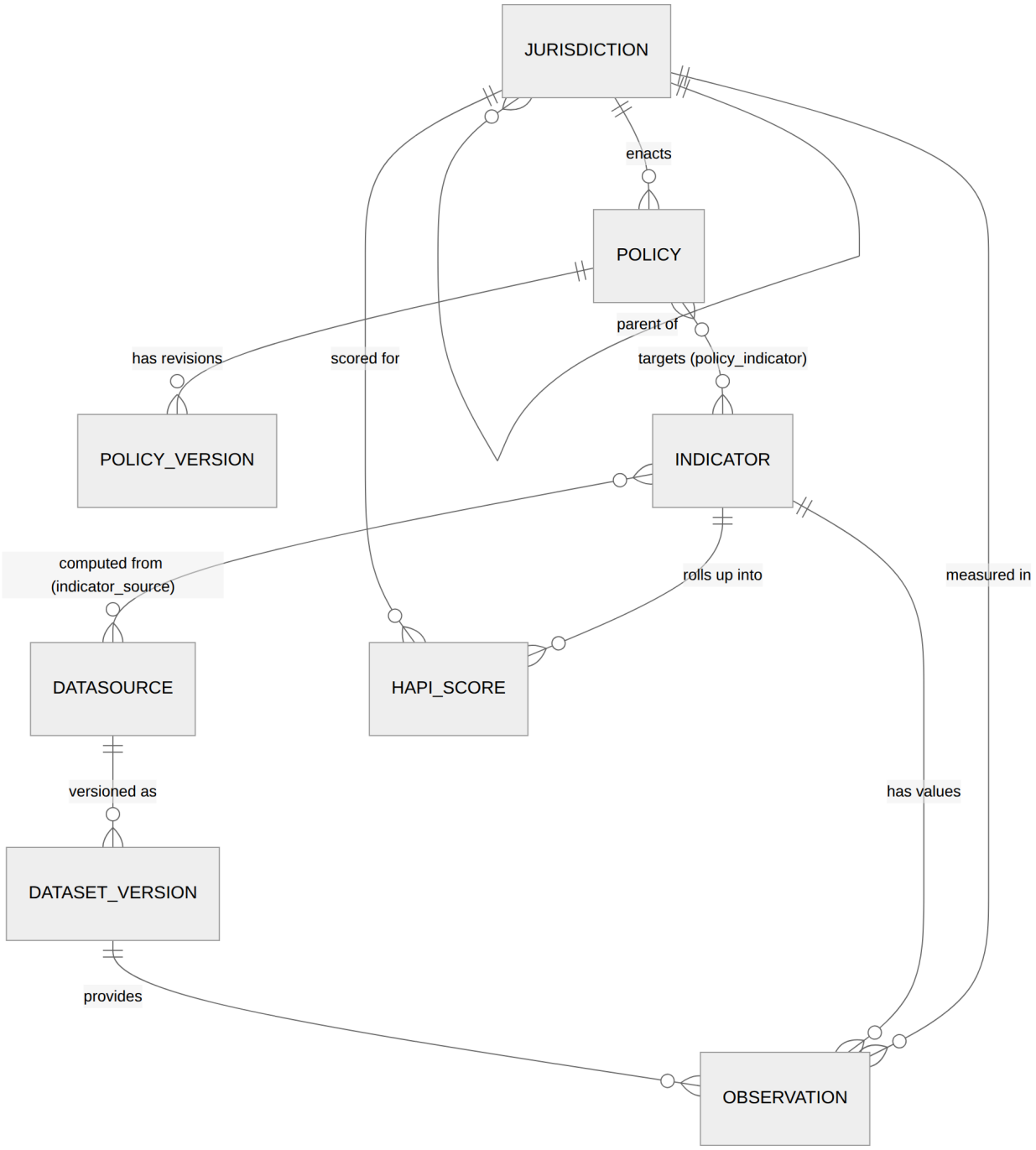


Figure 4. Entity–relationship model. Every value resolves along the lineage chain Observation → DatasetVersion → DataSource; indicators roll up into HAPI scores. (Source: `docs/03-data-model.md`.)

6. Instantiating the cycle

To show the framework runs end-to-end, the Observatory instantiates its three analytical stages. We summarize each here as evidence; the full methodology of each is the subject of its own paper (§8).

Indicator — HAPI. The Healthy Aging Policy Index scores how well a jurisdiction supports healthy aging (0–100) across six domains (Health, Independence, Social Participation, Financial Security, Care Access, Digital Inclusion). Each indicator declares a direction, a min-max normalization against a documented reference range, and, for counts, a per-capita rule; the composite is a weighted mean over available domains, renormalized by the domains present. Crucially, weighting — the most contestable step — is treated as an auditable object: three schemes (equal; a theory-anchored *expert* tier from the WHO [1] and AgeWatch [4] frameworks; and an *empirical*, coefficient-of-variation scheme) are reported side by side per OECD/JRC guidance [3], and every score carries a `method_version`. *The index's methodology, validation, and sensitivity analysis are developed fully in Paper 2.*

Outcome — analytics under a causal discipline. Results are two-tier and always labelled. **Tier 1 (Association)**: trends, policy-event overlays, correlations. **Tier 2 (Causal)**: quasi-experimental designs — interrupted time series (ITS), difference-in-differences, synthetic control — each carrying its assumptions as attached metadata. The reference Tier-2 design is segmented-regression ITS [7, 8] with Newey–West (HAC) inference [6]; it refuses to report coefficients below three points per segment, returning an explicit `insufficient_data` status instead. A finding is tagged `Causal` *only* together with its named assumptions, never upgraded silently from an association. *The empirical evaluations are the subject of Paper 4.*

Feedback — grounded AI assistant. Given a topic, the assistant retrieves an **evidence pack** (policies `P#`, literature `L#`, findings `F#`) from the platform's own stores, then drafts a review *from that exact pack* under a contract: every claim ends with a citation tag resolving to a pack item; only the pack may be used; analytic tiers are preserved, never upgraded; the output is a draft for a human, not a verdict. Because the pack is fixed and visible, a fabricated citation is structurally detectable, and every generation is logged. *The grounded-AI framework is developed in Paper 3.*

7. Design evaluation

We evaluate the design against the four research questions, using the instrument instantiated on Nova Scotia (CA-NS) and the federal level (CA) with a Care Access focus. All results are from the production instance, which runs the connectors against live upstream sources, as of 11 October 2026, generated by `ingest → score → analyze → paper-tables`. The instantiated corpus:

Asset	Count
Policies	26
Indicators (distinct, observed)	37
Observations	1,263
Dataset versions	47
Data sources	18
HAPI scores	174
Analytic findings	17
Literature references	7
Jurisdictions	3

Observations are append-only: when an upstream table is revised, the new dataset version's observations are stored alongside the superseded version's, so the count spans all 47 versions.

RQ1 — Provenance: can every number be traced? Yes. Every one of the 1,263 observations is bound to a DatasetVersion (47) and thence a DataSource (18); every HAPI score and

figure is computed from those observations; and the Data Hub renders the full `Observation → DatasetVersion → DataSource` chain for each value. Traceability is not a sampled audit but a structural invariant of the schema (§5): a value that lacks lineage cannot be stored. *Result: 100% of published quantitative values are source-traceable.*

RQ2 — Reproducibility: does the pipeline reproduce, idempotently? Yes. The transforms are deterministic, so the same inputs yield identical scores and findings; re-ingesting unchanged upstream data is a no-op because dataset versions are content-addressed by checksum. A demonstration instance regenerates from committed fixtures (captured from the same upstream tables) with a single command; the production numbers above regenerate from the stored dataset versions with `paper-tables`. *Result: deterministic reproduction; idempotent ingestion.*

RQ3 — Grounding: can the AI emit an ungrounded citation? No, by construction. The assistant may cite only ids present in the pre-assembled evidence pack; any tag that does not resolve to a pack item is detectable as fabrication, and every generation is logged for audit. Grounding is therefore a property of the retrieval-then-constrain design rather than of model behavior. *Result: zero ungrounded citations are representable; violations are structurally detectable.*

RQ4 — Robustness: are index comparisons robust to weighting? Partly — and the instrument says so. The six domains are measured on different survey vintages, so the test compares each jurisdiction's full domain profile (its latest available score in every domain) under the three schemes:

Jurisdiction	Domains	Data years	equal	expert	empirical
CA (federal)	6	2022–2025	53.9	52.7	51.6
CA-NS (Nova Scotia)	6	2022–2026	51.1	50.1	51.7

Composite *levels* are stable: the **maximum spread across schemes is 2.3 points**. The *ordering* is not: the federal composite is higher under the equal and expert schemes, Nova Scotia's under the empirical scheme (by 0.1 points). Two jurisdictions this close cannot be ranked by the index, and the sensitivity analysis is what shows it — a single-scheme dashboard would have published a ranking. Testing at a single period would have hidden the problem: the newest period holds one domain only (Care Access), for which every scheme returns the same value, an apparent spread of zero. For completeness, the computed v1 domain profiles are:

Jur.	Composite	Health	Indep.	Social	Financial	Care	Digital
CA (fed.)	53	53	53	44	59	42	72
CA-NS	50	41	47	55	50	54	59

Jur.: CA = federal, CA-NS = Nova Scotia. Columns are the six HAPI domains — Health, Indep(endence), Social (Participation), Financial (Security), Care (Access), Digital (Inclusion) — each showing its latest available score; Composite = expert-weighted mean of the six; 0–100, higher is better.

Causal discipline in action. The three ITS designs illustrate the Outcome-stage guarantee: each reports coefficients only because both segments meet the three-point minimum, and each is tagged **Causal(ITS)** only alongside its four assumptions. Two are attached to real NS policies. Around the 2022 long-term-care capital plan (12 pre / 4 post points), the waitlist shows a level *increase*

(+614.7; 95% CI 147.5..1081.9; p = 0.010; $R^2 = 0.26$). Around the 2022 Continuing Care Assistant workforce strategy (22/3, at the minimum), workforce per 1,000 seniors shows a level drop (−728.2; 95% CI −1441.2..−15.2; p = 0.045) and a steeper slope (+84.5; 95% CI 29.9..139.1; p = 0.002). A third, illustrative series (5/5) shows a downward slope change (−715.7; 95% CI −1125.1..−306.3; p < 0.001; $R^2 = 0.934$). These are design demonstrations, not evaluations: a few post-period points, a COVID-era baseline, and coincident system changes all strain the no-coincident-shock assumption, which is exactly why the assumptions travel with the estimate. The evaluations are Paper 4's task.

Table 1 summarizes the four results.

Table 1. Design-evaluation results.

RQ	Criterion	Evidence	Result
RQ1	Traceability	1,263 observations bound to lineage	100% source-traceable (structural invariant)
RQ2	Reproducibility	deterministic pipeline; content-addressed versions	identical re-runs; idempotent ingestion
RQ3	Grounding	fixed, visible evidence pack	0 ungrounded citations representable
RQ4	Robustness	3 weighting schemes on full domain profiles	levels stable (max spread 2.3 pts); CA–NS ordering not robust, reported as such

Together these answer the design questions: the observatory is traceable, reproducible, and grounded, and its robustness check works as intended — it bounds how far the weighting can move the index and declines to rank jurisdictions the weighting can reorder. These are the properties the Intelligence Cycle (§3) identifies as where trust is won or lost.

8. The research program

Scoping Paper 1 to the observatory design lets each analytical module be developed as a focused study that both deepens and reuses the platform:

Paper	Contribution	Instantiates
1 (this paper)	Design of a reproducible healthy-aging policy observatory + the Policy Intelligence Cycle	Research infrastructure
2	The HAPI methodology — a transparent, sensitivity-tested composite-indicator framework	Indicator stage
3	AI-assisted policy intelligence — grounded retrieval + evidence synthesis	Feedback stage
4	Causal evaluation of Canadian aging policies — ITS / DiD / synthetic control	Outcome stage

Each paper deposits assets back into the platform (a curated corpus, a validated index, reusable agent tooling, worked evaluations), so the program compounds rather than restarts — the strategic point of building an instrument rather than a study.

9. Limitations and ethics

Limitations. (1) *Coverage — an initial implementation.* This paper evaluates the *design* on a deliberately small slice: two jurisdictions and a demonstration-scale corpus (26 policies, 1,263 observations). The empirical weighting and any cross-jurisdiction inference are therefore indicative — RQ4 shows two jurisdictions whose ordering the weighting can reverse — and the results should not be read as a mature observatory. The design's value is in the method and its additive extensibility: adding a province is a row insert plus one connector, so scaling is future work the architecture enables rather than a redesign. (2) *Normalization ranges are choices.* Min-max reference ranges are documented and contestable; RQ4's robustness check mitigates but does not eliminate this — a composite index is a lens, not a measurement. The composite also combines domains measured in different years (2022–2026), so it describes a recent profile rather than a single year. (3) *Causal identification.* ITS and related designs rest on assumptions (no coincident shocks, correct functional form) that real timelines strain; thin data may permit only *demonstrating* a design, which the platform reports honestly. (4) *Design evaluation, not outcome evaluation.* We evaluate whether the instrument is trustworthy (RQ1–RQ4), not whether any policy improved health — the latter is Paper 4's task.

Ethics. The platform uses aggregate, published statistics, not individual records, and respects Canada's federal privacy law (PIPEDA) as a baseline. Most importantly, the design encodes an ethic of *epistemic honesty*: association is never dressed as causation, index choices are exposed for scrutiny, and AI output is grounded and cited by construction. For infrastructure that may inform decisions about a vulnerable population, that honesty is a safety property.

10. Conclusion

We have argued that the weaknesses of aging-policy evidence are infrastructure problems, not analysis problems, and presented the design of a reproducible observatory that treats them as such — organized by the Healthy Aging Policy Intelligence Cycle, a policy-level Learning Health System. Evaluated as a design, the instrument is traceable (100% of values source-linked), reproducible (deterministic, idempotent), grounded (no representable ungrounded citation), and honest about robustness (index levels move at most 2.3 points across weightings, and a ranking the weighting can reverse is reported as a tie, not published). The contribution is the design itself: a durable paradigm for trustworthy aging-policy evidence, on which a focused program of methodology, grounded-AI, and causal-evaluation papers builds.

Acknowledgements

This work uses publicly available data from Statistics Canada, the Canadian Institute for Health Information (CIHI), and the Nova Scotia Open Data portal; we thank these providers for open access to the underlying tables. The author received no specific funding for this work.

Data and code availability

The platform — schema, pipeline, methodology, and web application — is developed in the open at https://github.com/GiorgioHuang/aging-policy-lab. Every quantitative result reproduces from the pipeline commands in the paper's repository README.

11. References

Aging frameworks and composite indices.

1. World Health Organization. *World Report on Ageing and Health.* Geneva: WHO, 2015. ISBN 978-92-4-156504-2.
2. World Health Organization. *Decade of Healthy Ageing 2021–2030.* Geneva: WHO, 2020.
3. Nardo, M., Saisana, M., Saltelli, A., Tarantola, S., Hoffmann, A., & Giovannini, E. *Handbook on Constructing Composite Indicators: Methodology and User Guide.* Paris: OECD Publishing, 2008. doi:10.1787/9789264043466-en.
4. HelpAge International. *Global AgeWatch Index 2015: Insight Report.* London: HelpAge International, 2015.
5. Zaidi, A., Gasior, K., Hofmarcher, M. M., Lelkes, O., Marin, B., Rodrigues, R., Schmidt, A., Vanhuysse, P., & Zolyomi, E. *Active Ageing Index 2012: Concept, Methodology and Final Results.* Vienna: European Centre, 2013.

Quasi-experimental methods (interrupted time series).

6. Newey, W. K., & West, K. D. "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica*, 55(3), 703–708, 1987. doi:10.2307/1913610.
7. Wagner, A. K., Soumerai, S. B., Zhang, F., & Ross-Degnan, D. "Segmented regression analysis of interrupted time series studies in medication use research." *Journal of Clinical Pharmacy and Therapeutics*, 27(4), 299–309, 2002. doi:10.1046/j.1365-2710.2002.00430.x.
8. Lopez Bernal, J., Cummins, S., & Gasparrini, A. "Interrupted time series regression for the evaluation of public health interventions: a tutorial." *International Journal of Epidemiology*, 46(1), 348–355, 2017. doi:10.1093/ije/dyw098.

Learning health systems, reproducible infrastructure, and grounded generation.

9. Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., et al. "The FAIR Guiding Principles for scientific data management and stewardship." *Scientific Data*, 3, 160018, 2016. doi:10.1038/sdata.2016.18.
10. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." *NeurIPS* 33, 2020. arXiv:2005.11401.
11. Friedman, C. P., Wong, A. K., & Blumenthal, D. "Achieving a Nationwide Learning Health System." *Science Translational Medicine*, 2(57), 57cm29, 2010. doi:10.1126/scitranslmed.3001456.

Primary data sources.

12. Statistics Canada. Data accessed via the Web Data Service (WDS) `getFullTableDownloadCSV`, filtered to Canada and Nova Scotia at the reference periods each table publishes. Tables used (as cited in the Data Hub connectors): **13-10-0389** *Life expectancy, at birth and at age 65, by sex* (three-year averages; Health); **13-10-0971** *Health-adjusted life expectancy* (periodic; Health); **13-10-0096** *Health characteristics, annual estimates* (Canadian Community Health Survey; Health, Care Access, Social Participation); **13-10-0966** *Functional health* (CCHS cycles, 2015 onward; Independence); **13-10-0374** disability rate, 65+ (Canadian Survey on Disability; Independence); **13-10-0789** *Health characteristics of seniors aged 65 and over* (Canadian Health Survey on Seniors, 2019–2020; Independence — ADL); **11-10-0135** *Low income statistics by age, gender and economic family type* (annual; Financial Security);

14-10-0202 *Employment by industry, annual* (SEPH, 2001–, NAICS 623; Care Access workforce); **22-10-0135** *Internet use* (Canadian Internet Use Survey; Digital Inclusion); and **17-10-0005** *Population estimates on July 1, by age and gender* (per-capita denominators). Ottawa: Statistics Canada.

13. Canadian Institute for Health Information (CIHI). Long-term-care and continuing-care data holdings, transitioning to the Integrated interRAI Reporting System (IRRS) from the legacy Home Care Reporting System (HCRS) and Continuing Care Reporting System (CCRS); interRAI Home Care (RAI-HC) and MDS 2.0 assessment instruments. Used here for the LTC bed-capacity and caregiver-distress signals. Ottawa: CIHI.
14. Government of Nova Scotia. Nova Scotia Open Data Portal — including the LTC placement waitlist (c39g-gsdd) and the LTC facilities directory (x76a-axw2). data.novascotia.ca.